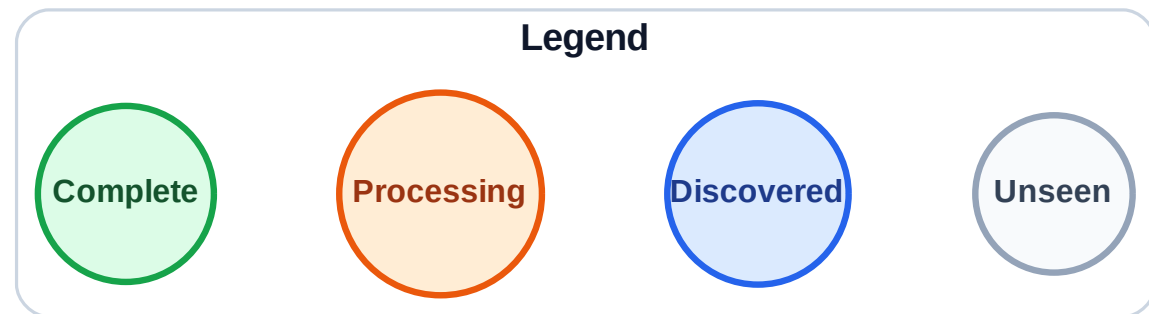
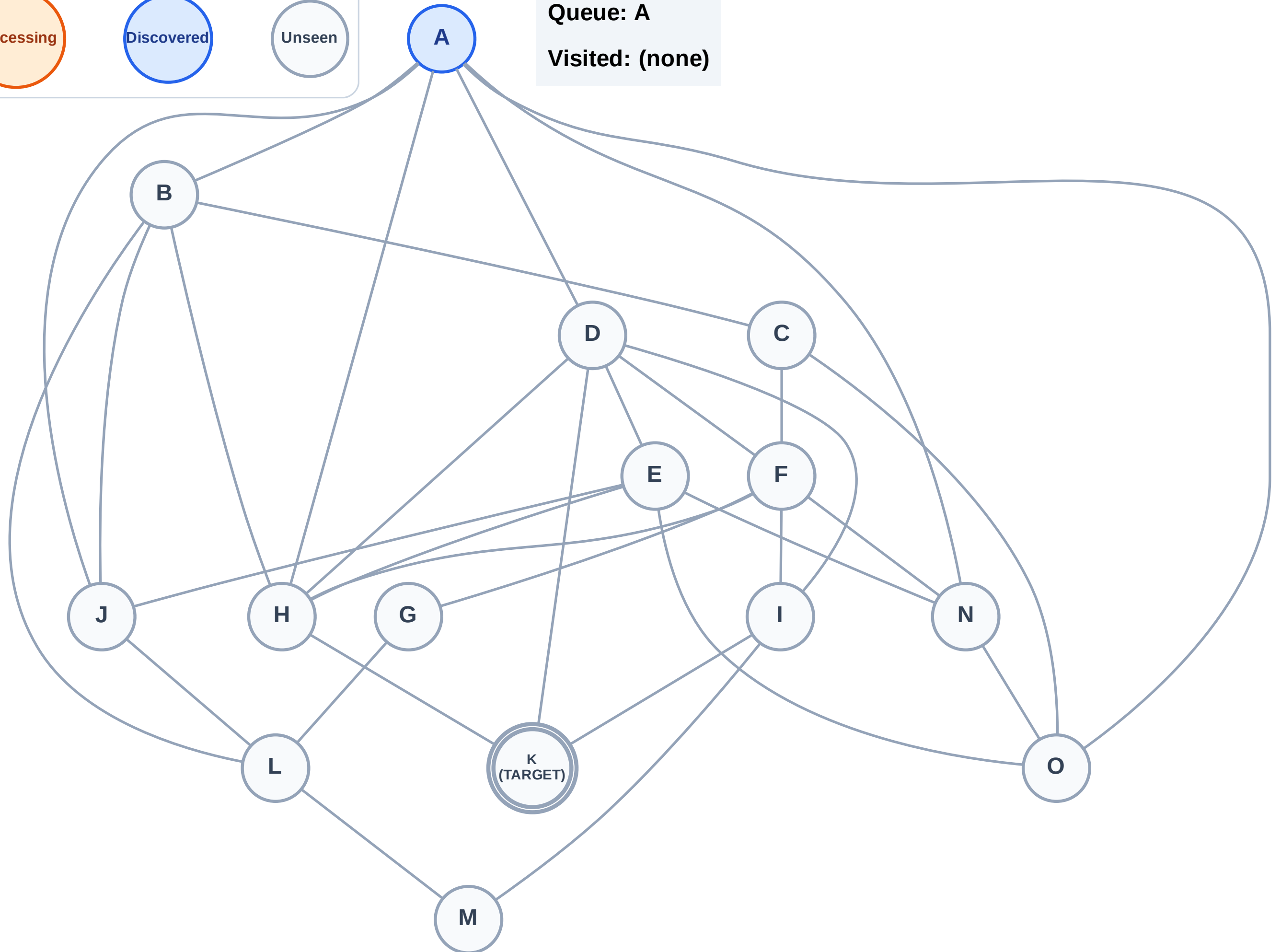


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

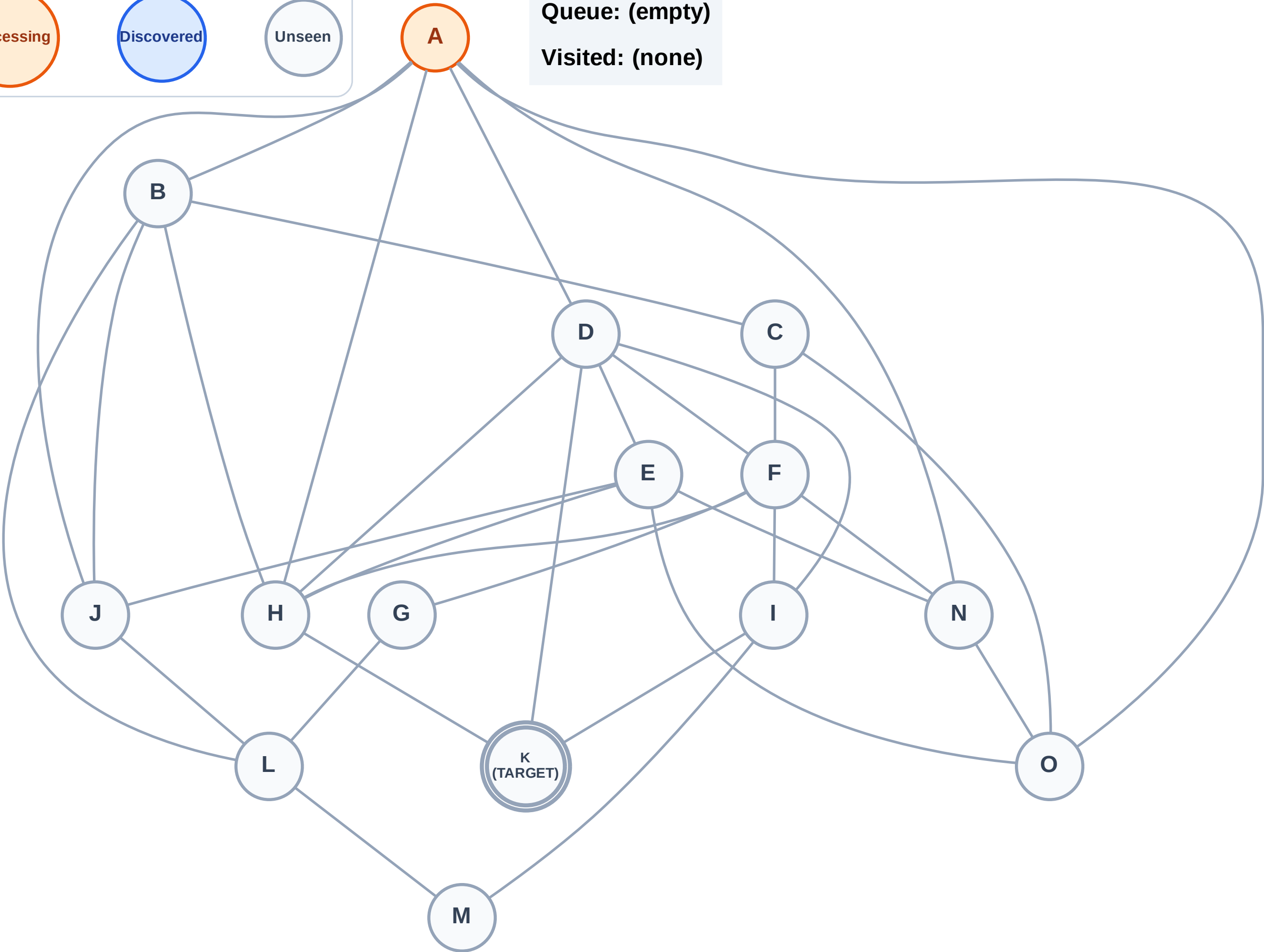
Processing

Discovered

Unseen

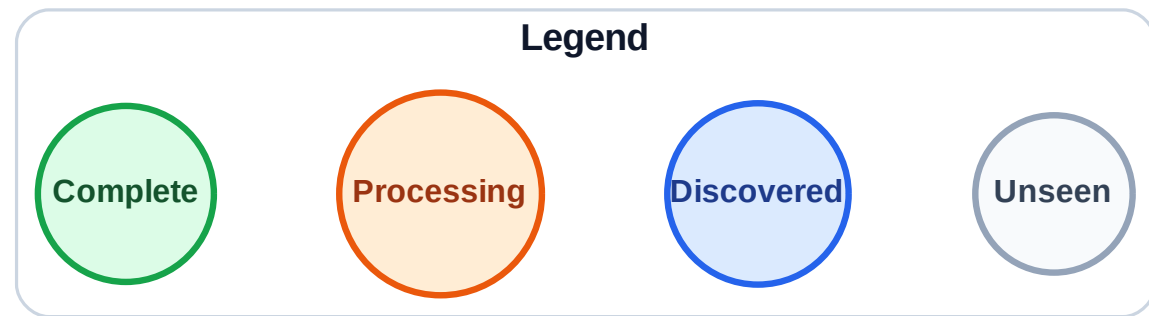
Queue: (empty)

Visited: (none)

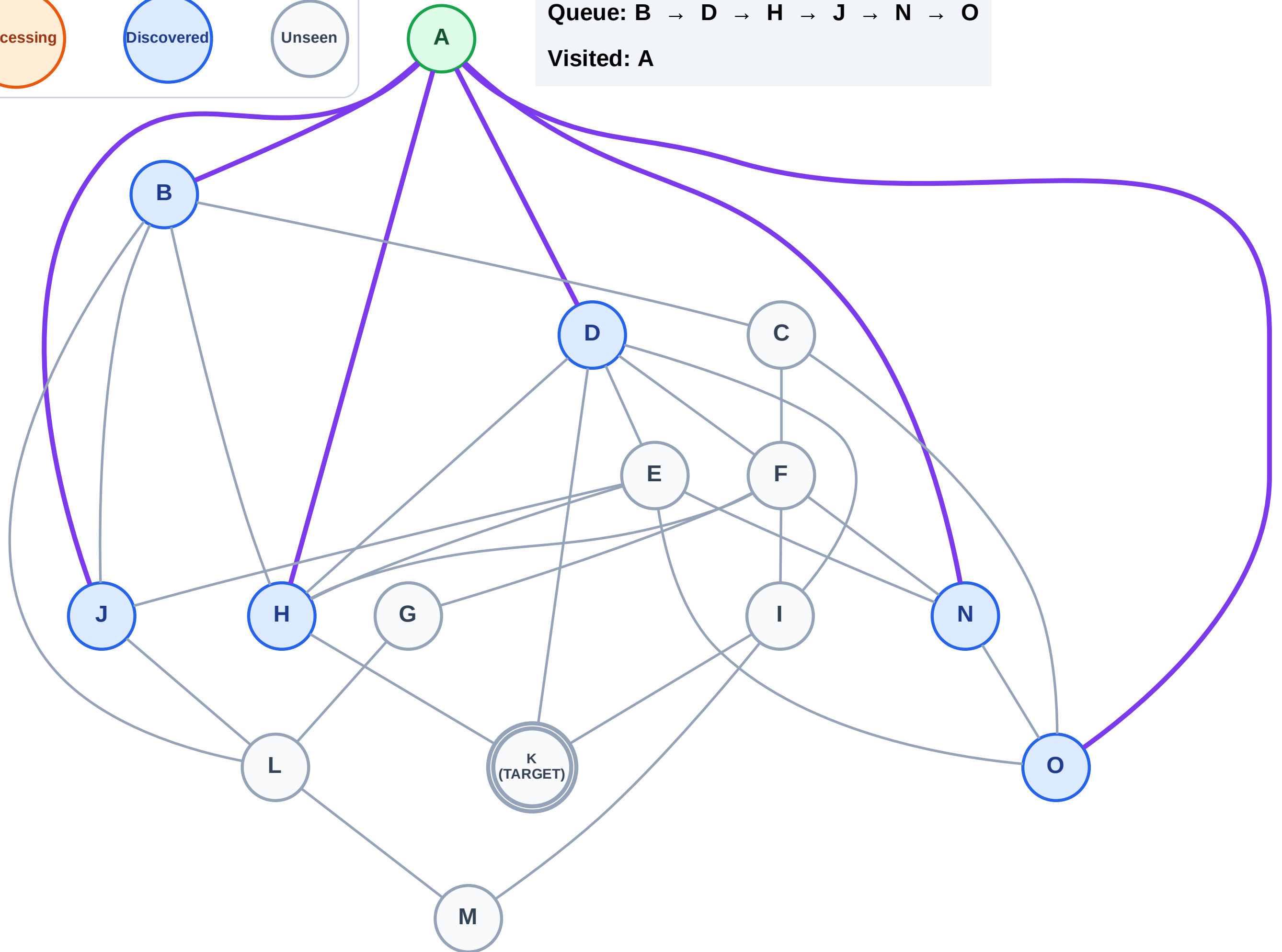


BFS Traversal

Step 3: Discover B, D, H, J, N, O from A

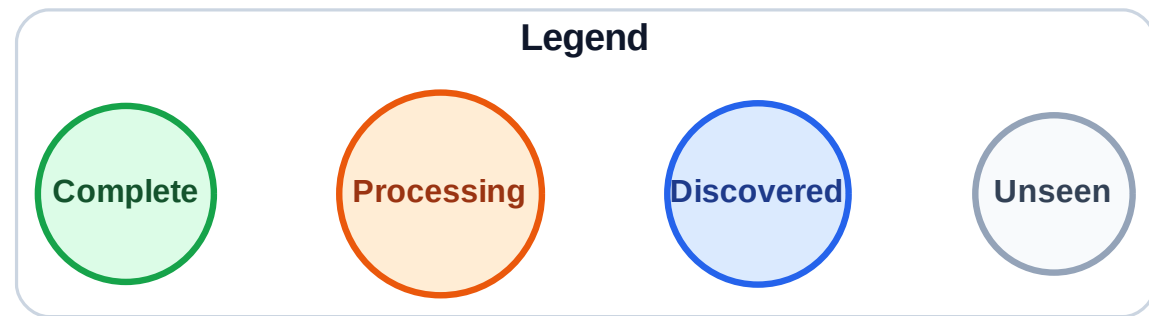


Queue: B → D → H → J → N → O
Visited: A



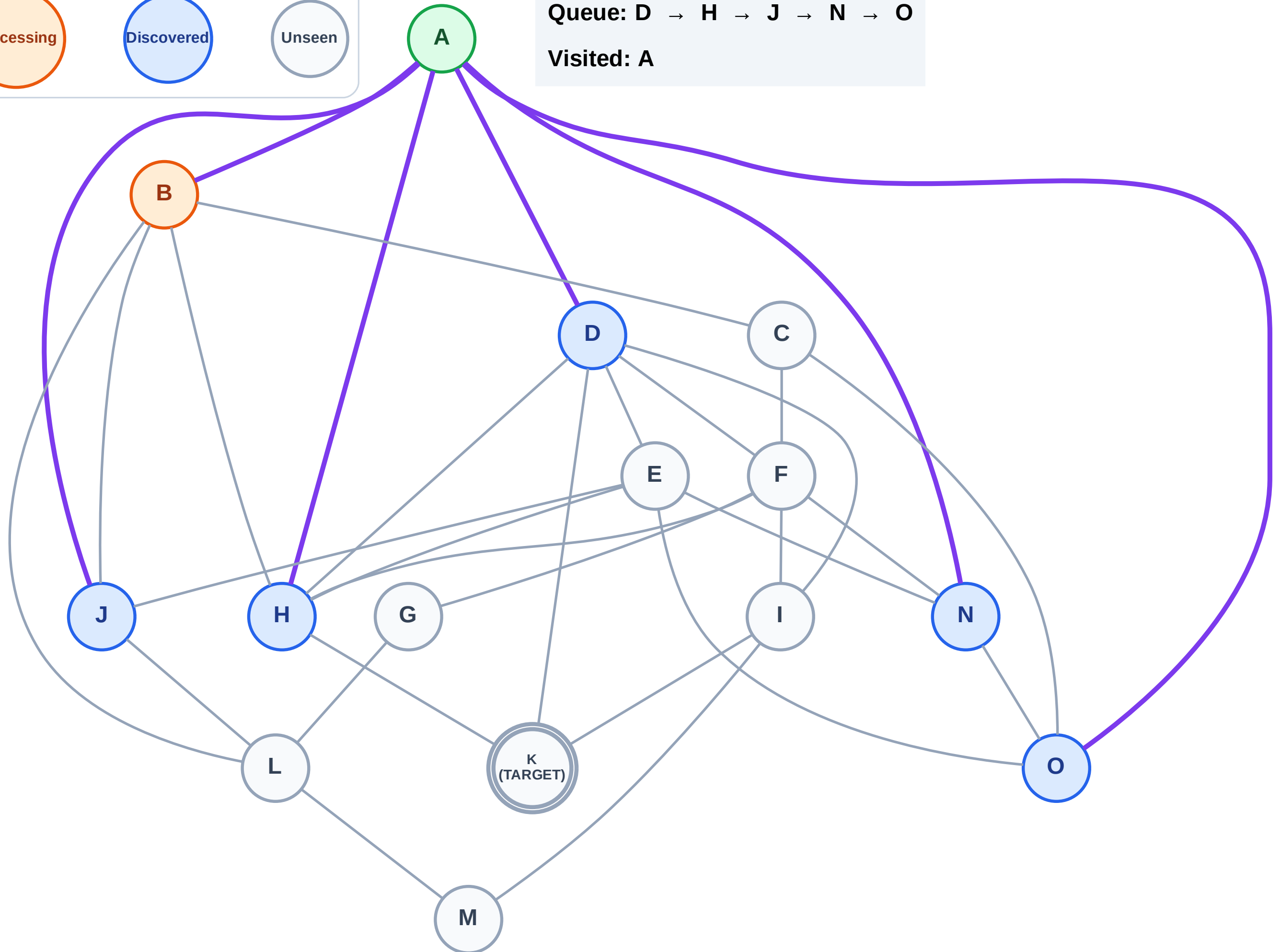
BFS Traversal

Step 4: Process node B



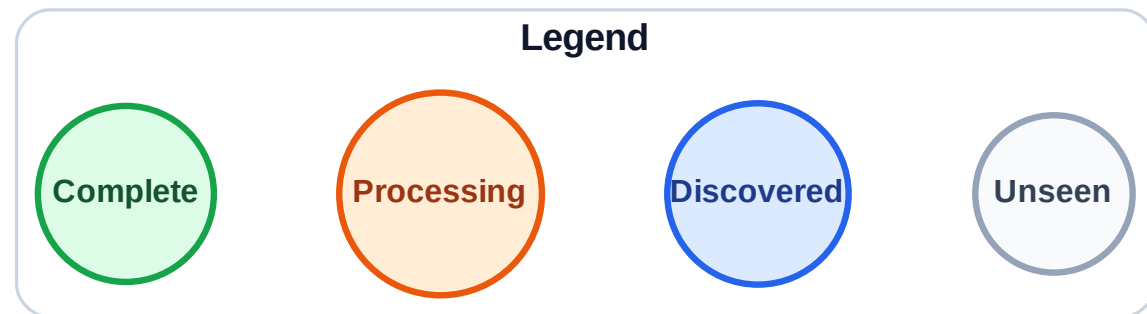
Queue: D → H → J → N → O

Visited: A



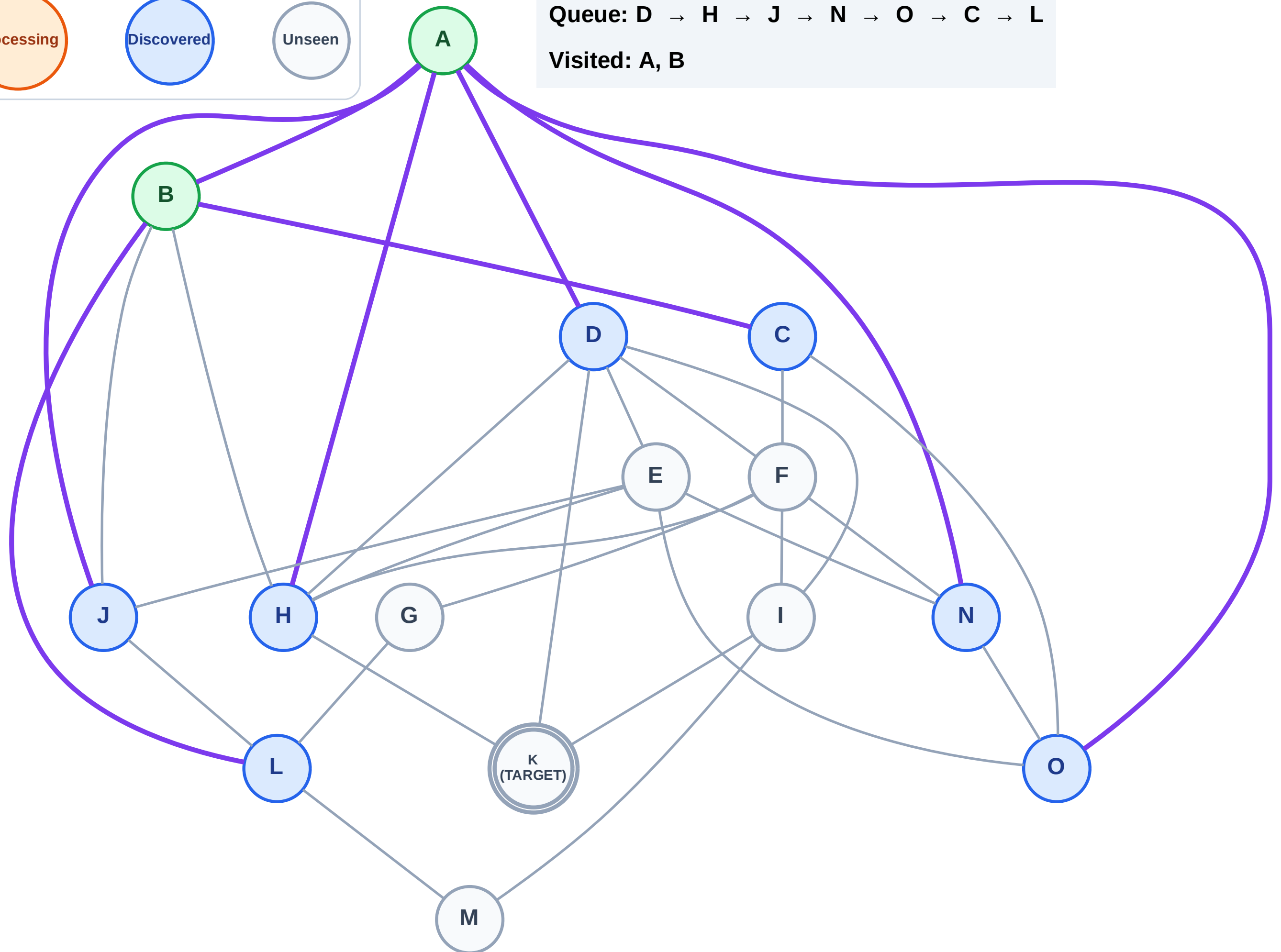
BFS Traversal

Step 5: Discover C, L from B



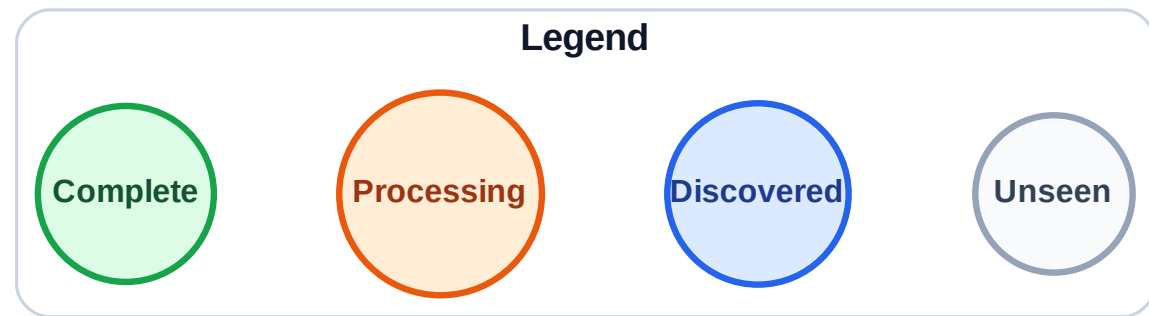
Queue: D → H → J → N → O → C → L

Visited: A, B



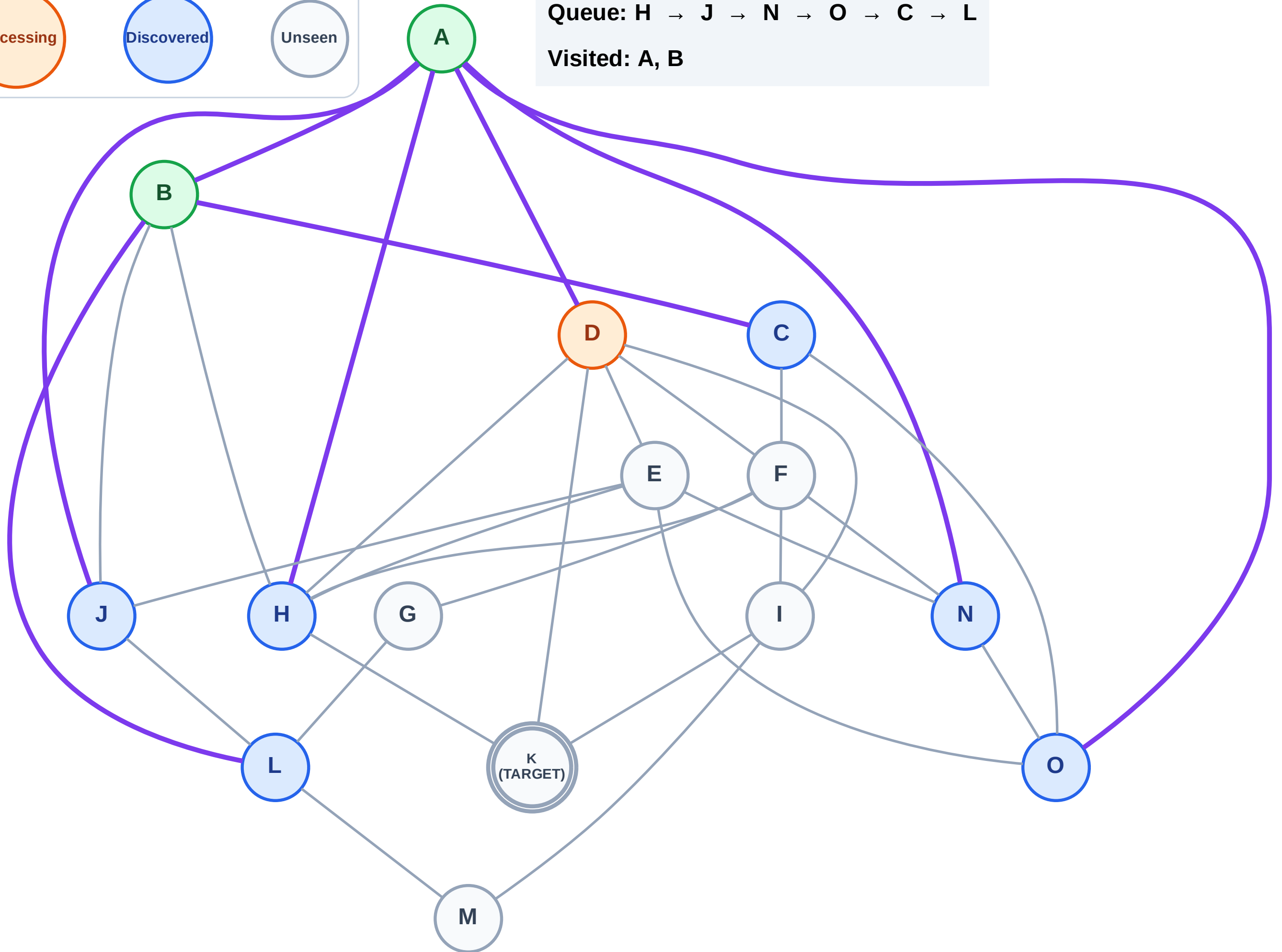
BFS Traversal

Step 6: Process node D



Queue: H → J → N → O → C → L

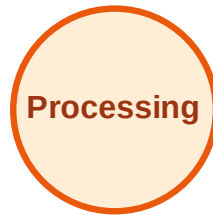
Visited: A, B



BFS Traversal

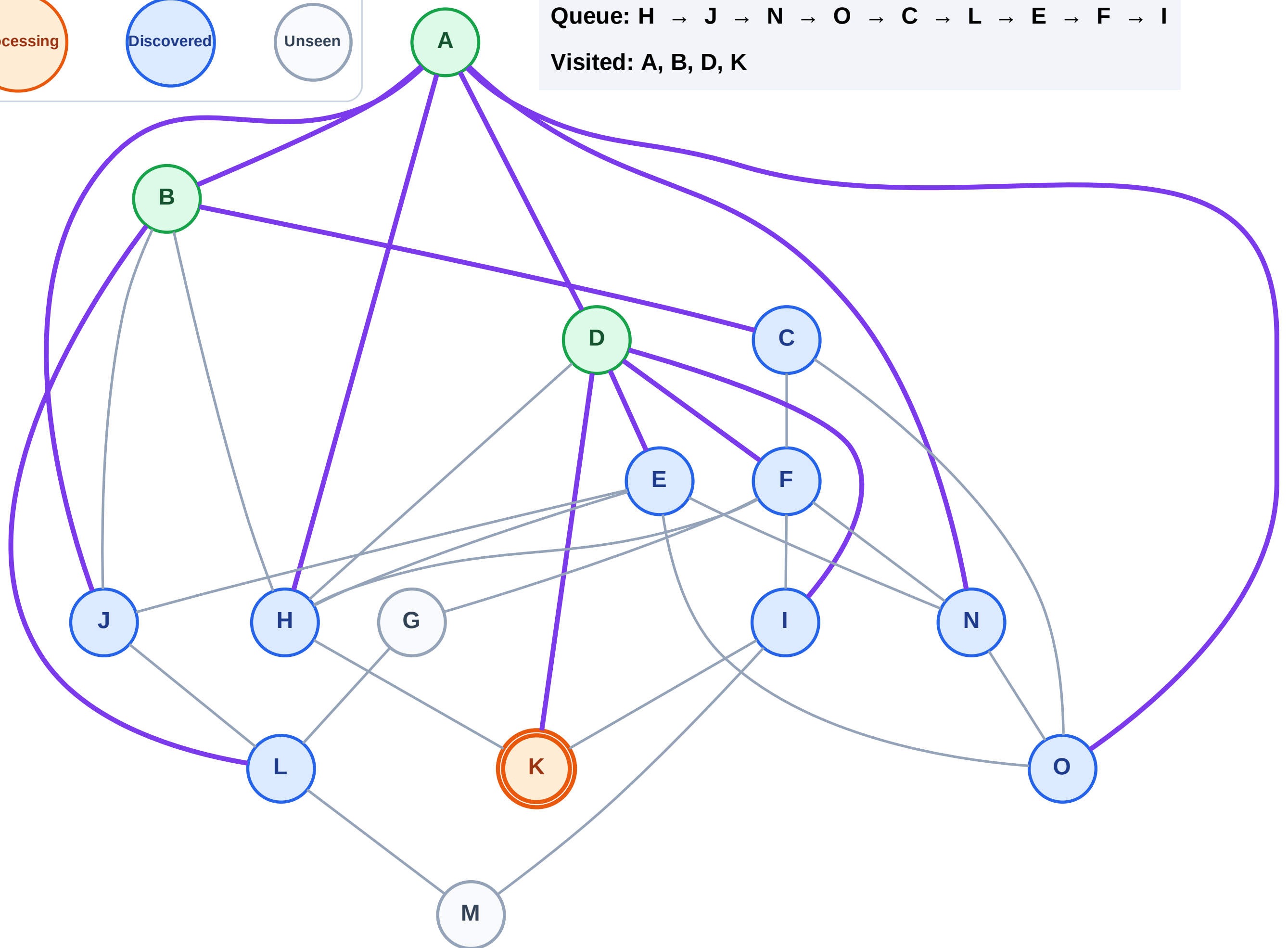
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend



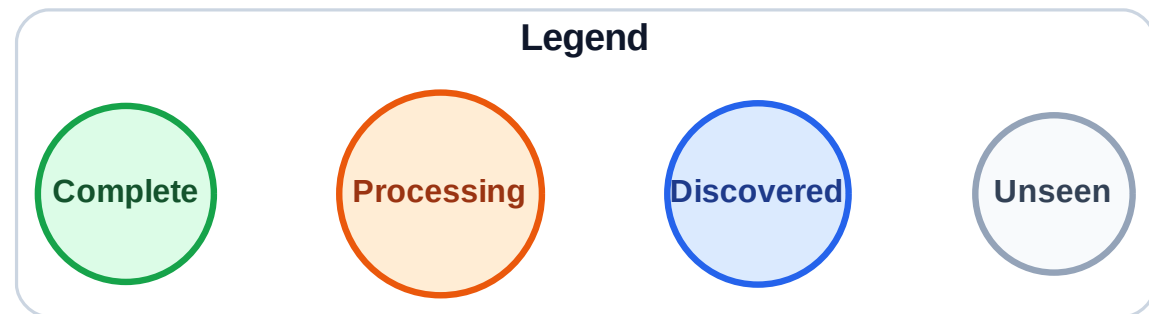
Queue: H → J → N → O → C → L → E → F → I

Visited: A, B, D, K

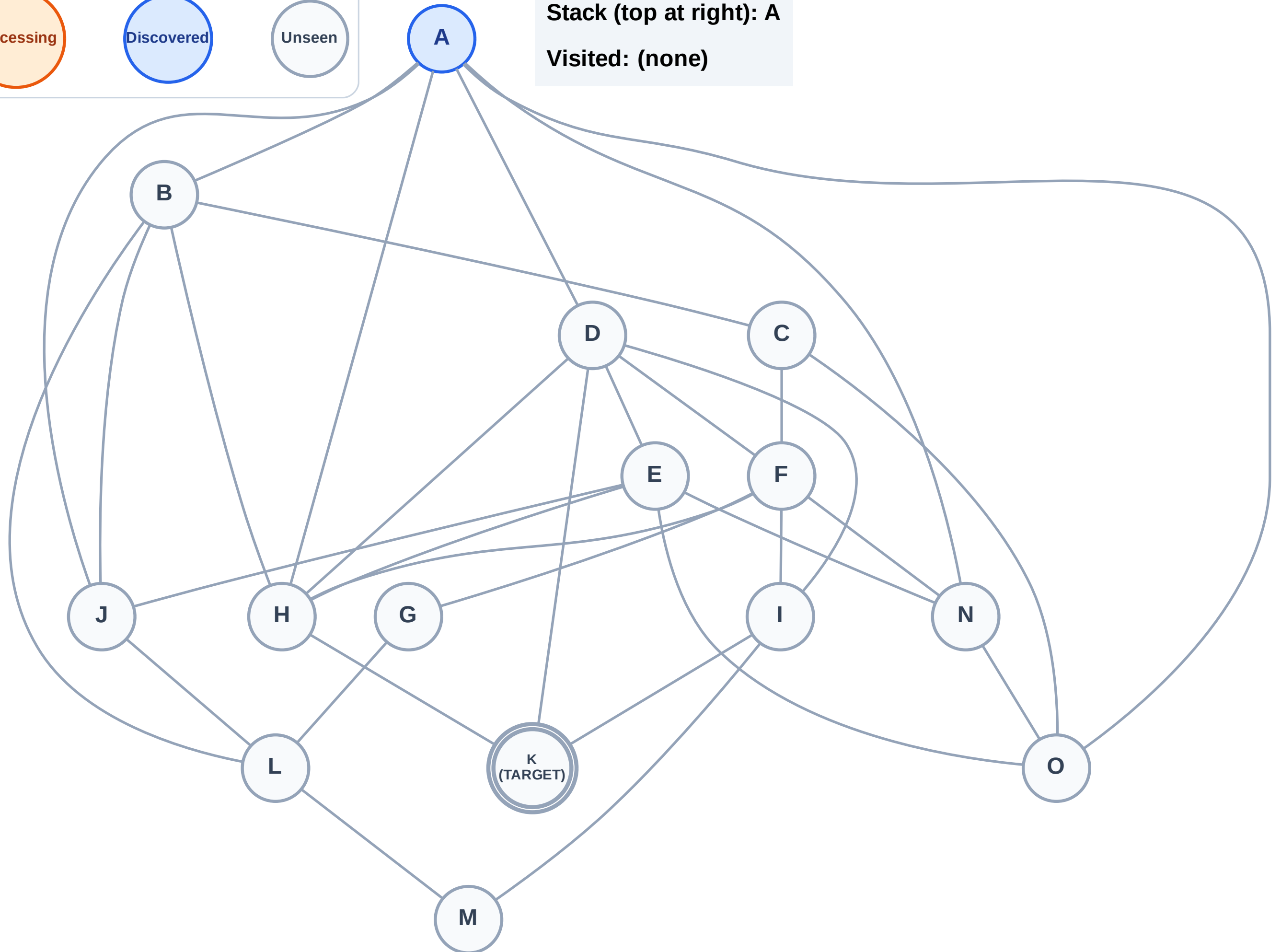


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A
Visited: (none)



DFS Traversal

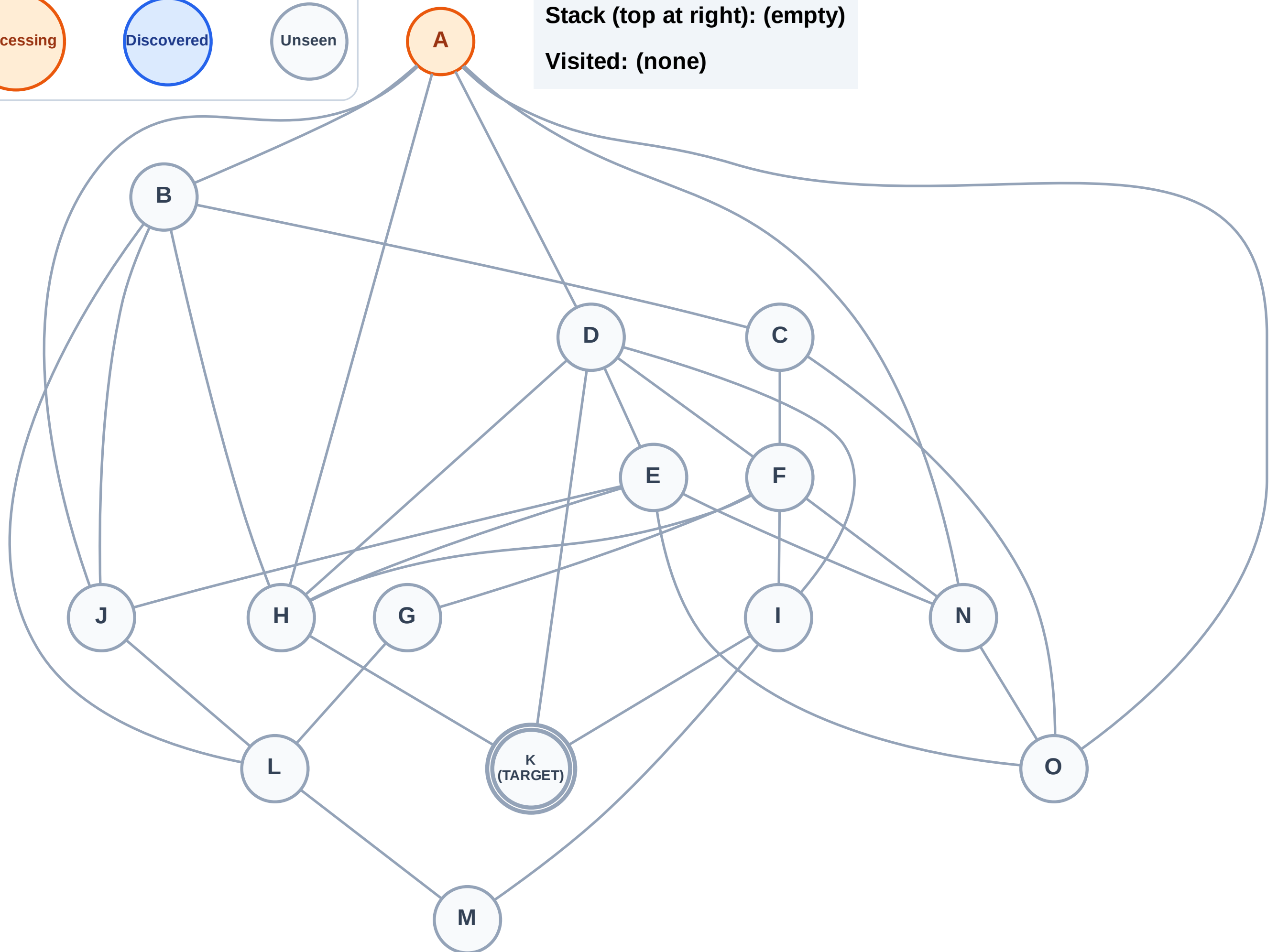
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)

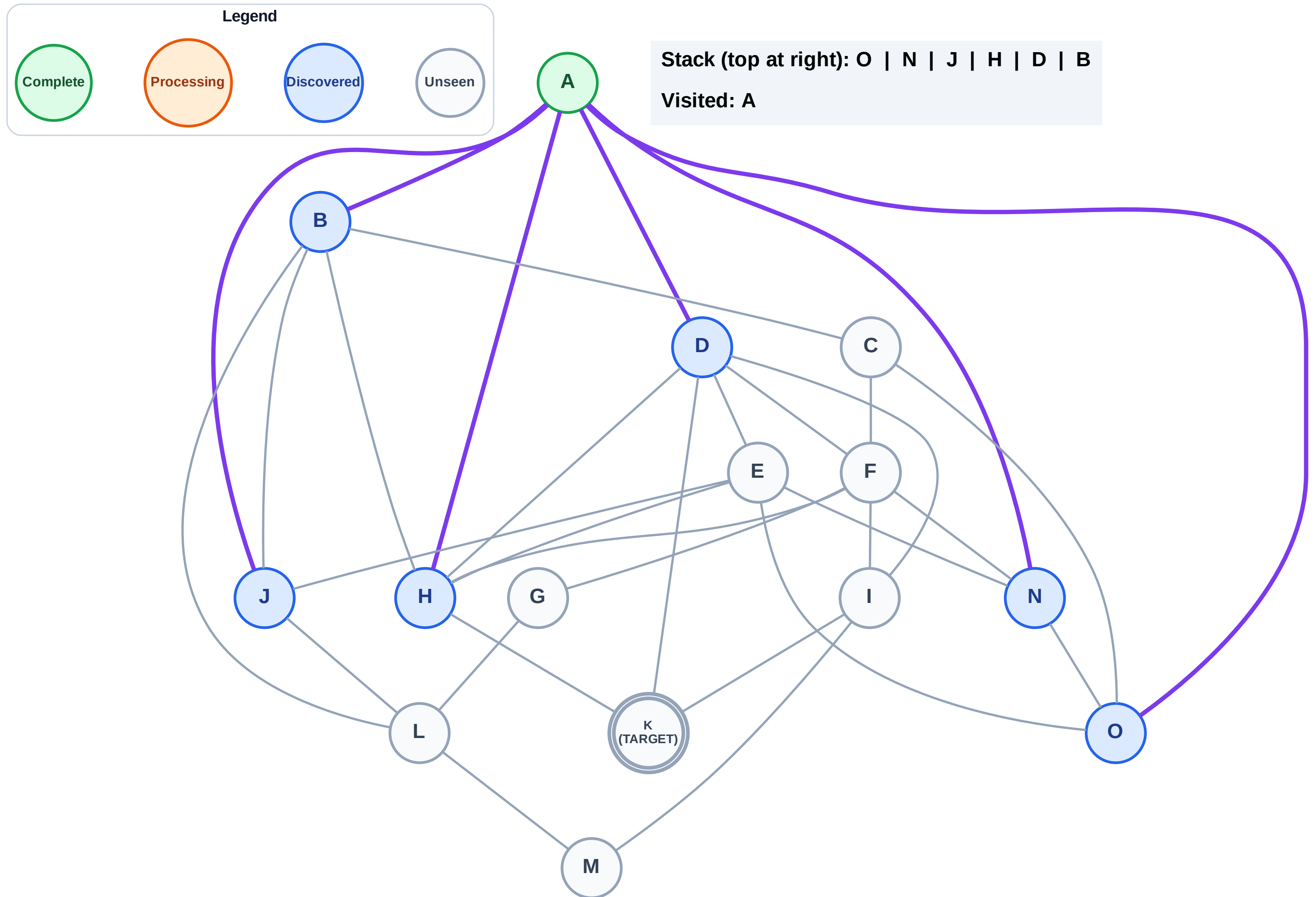


DFS Traversal

Step 3: Discover O, N, J, H, D, B from A

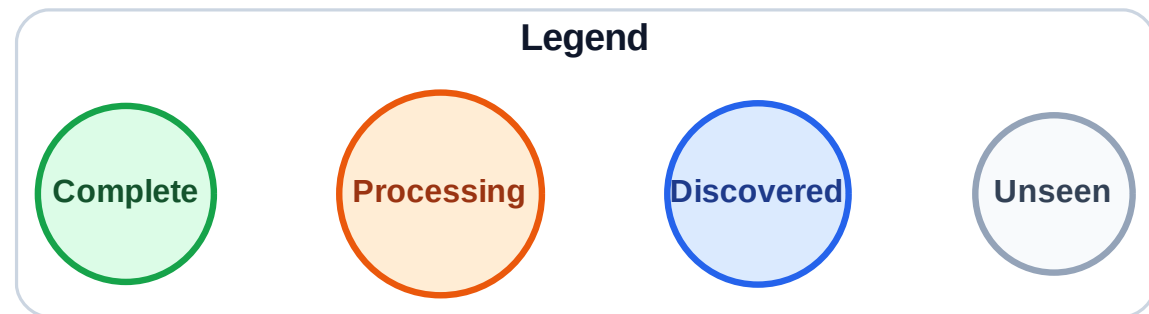
DFS Traversal

Step 3: Discover O, N, J, H, D, B from A

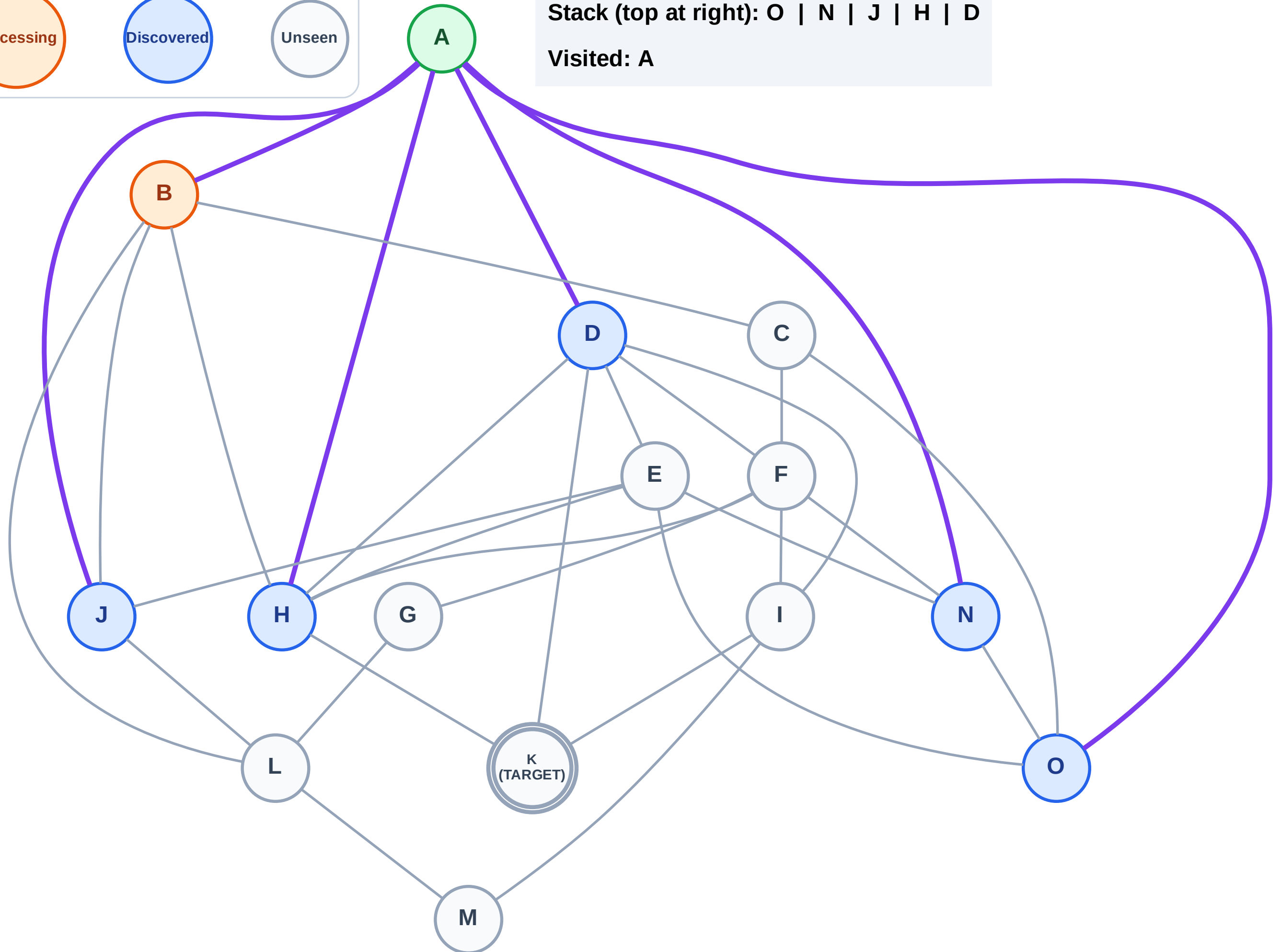


DFS Traversal

Step 4: Process node B

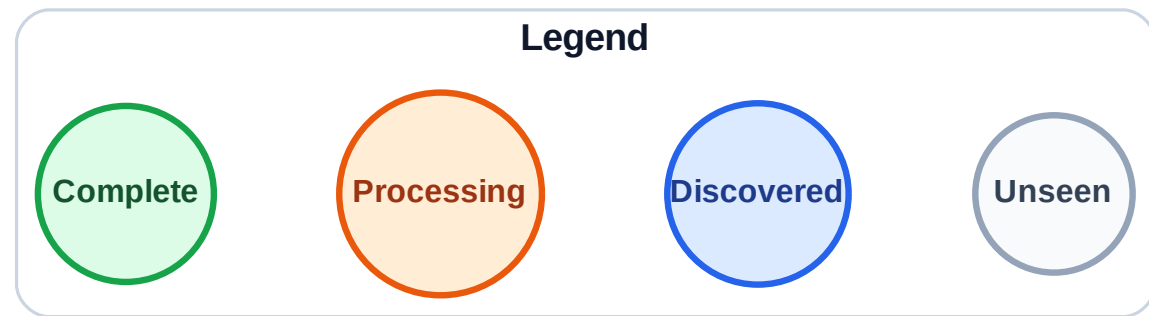


Stack (top at right): O | N | J | H | D
Visited: A



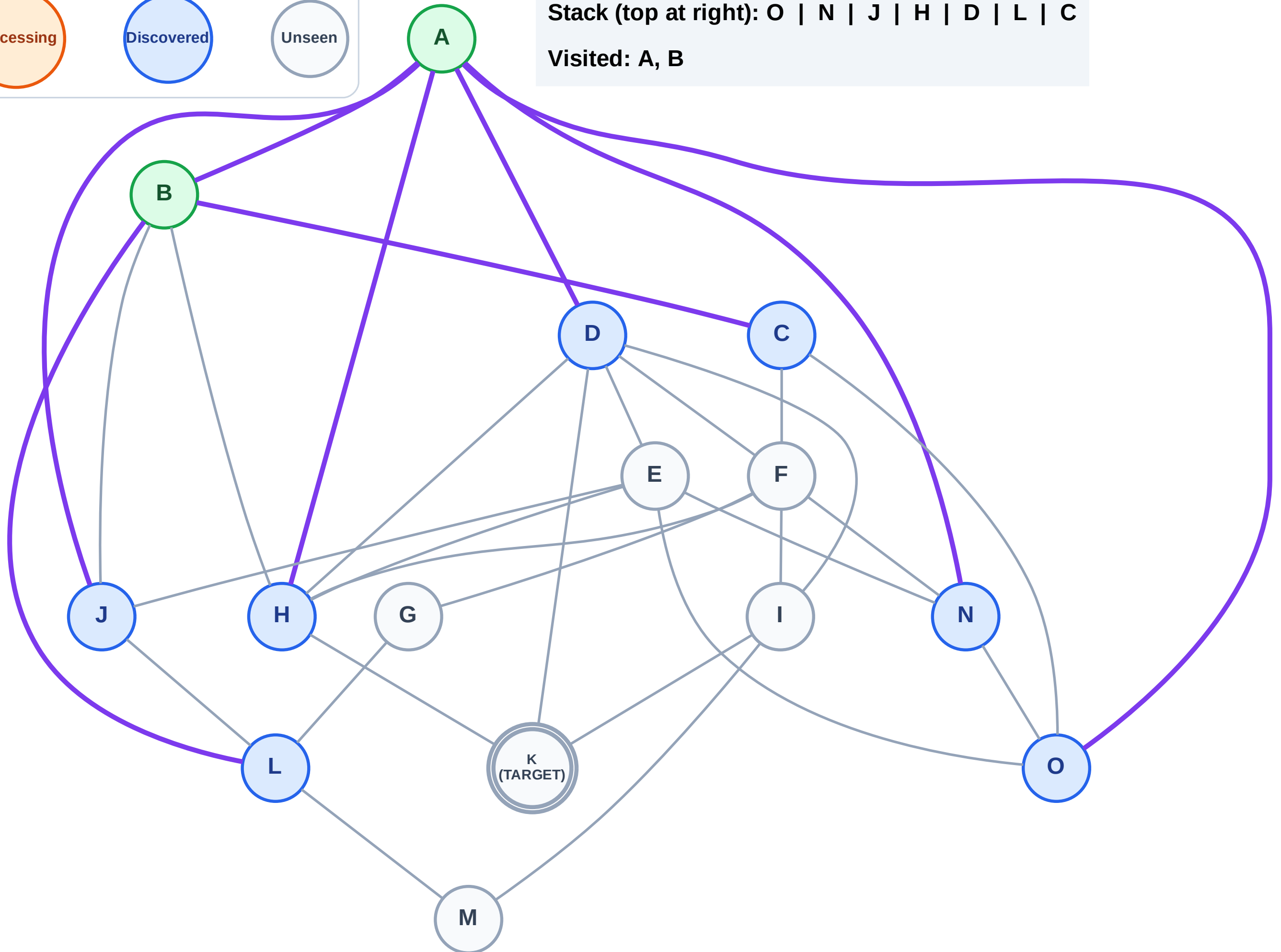
DFS Traversal

Step 5: Discover L, C from B



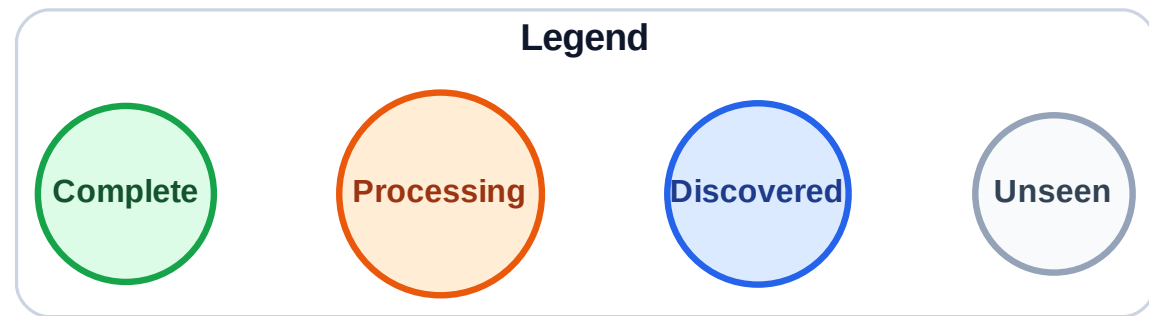
Stack (top at right): O | N | J | H | D | L | C

Visited: A, B



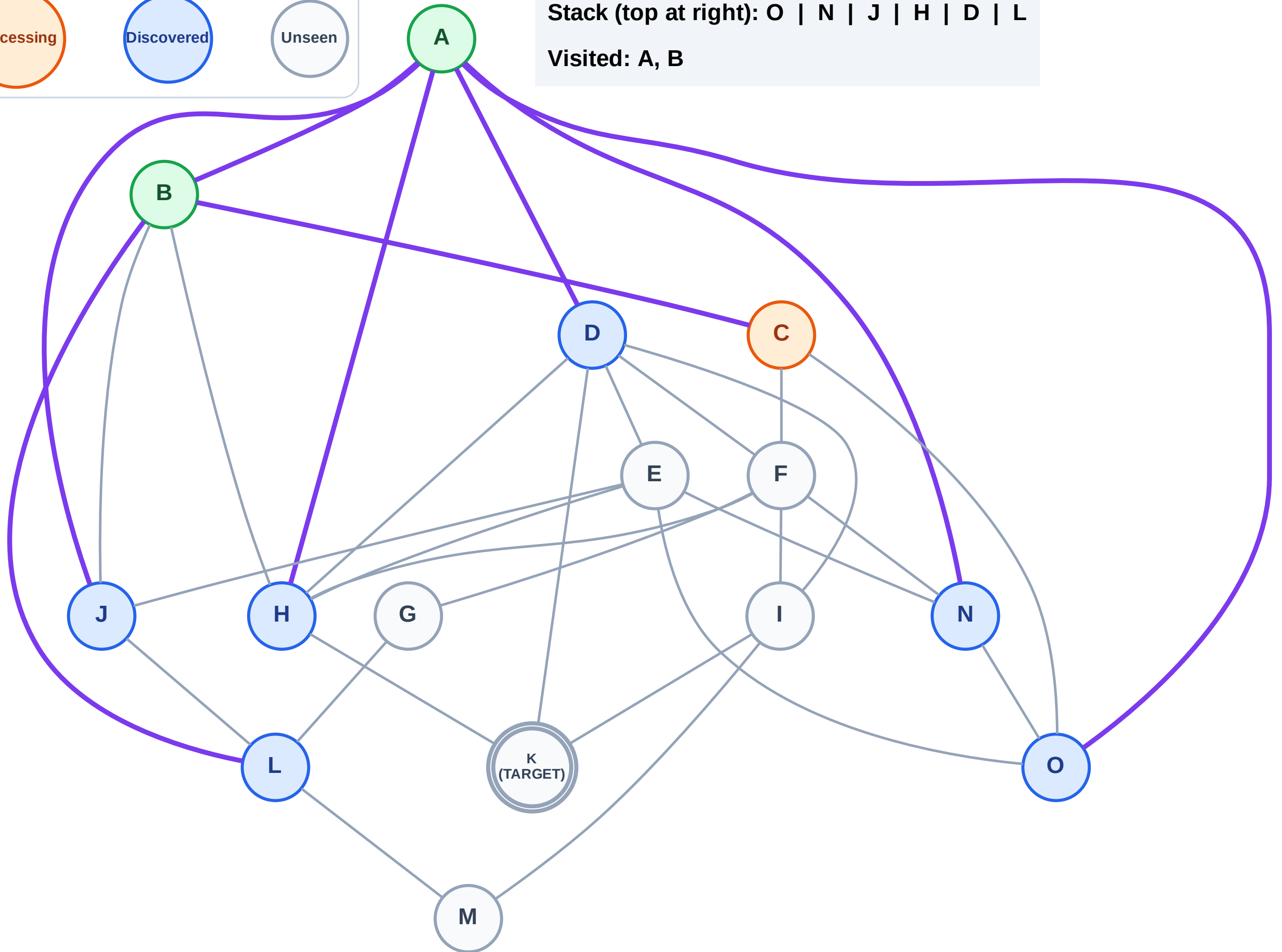
DFS Traversal

Step 6: Process node C



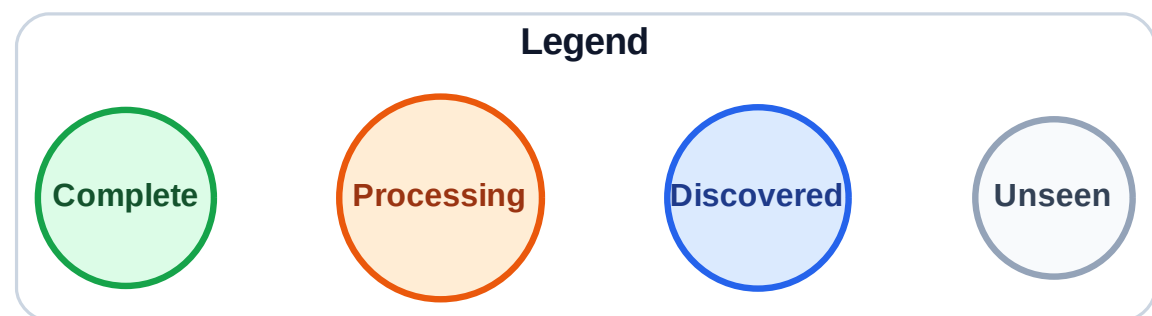
Stack (top at right): O | N | J | H | D | L

Visited: A, B



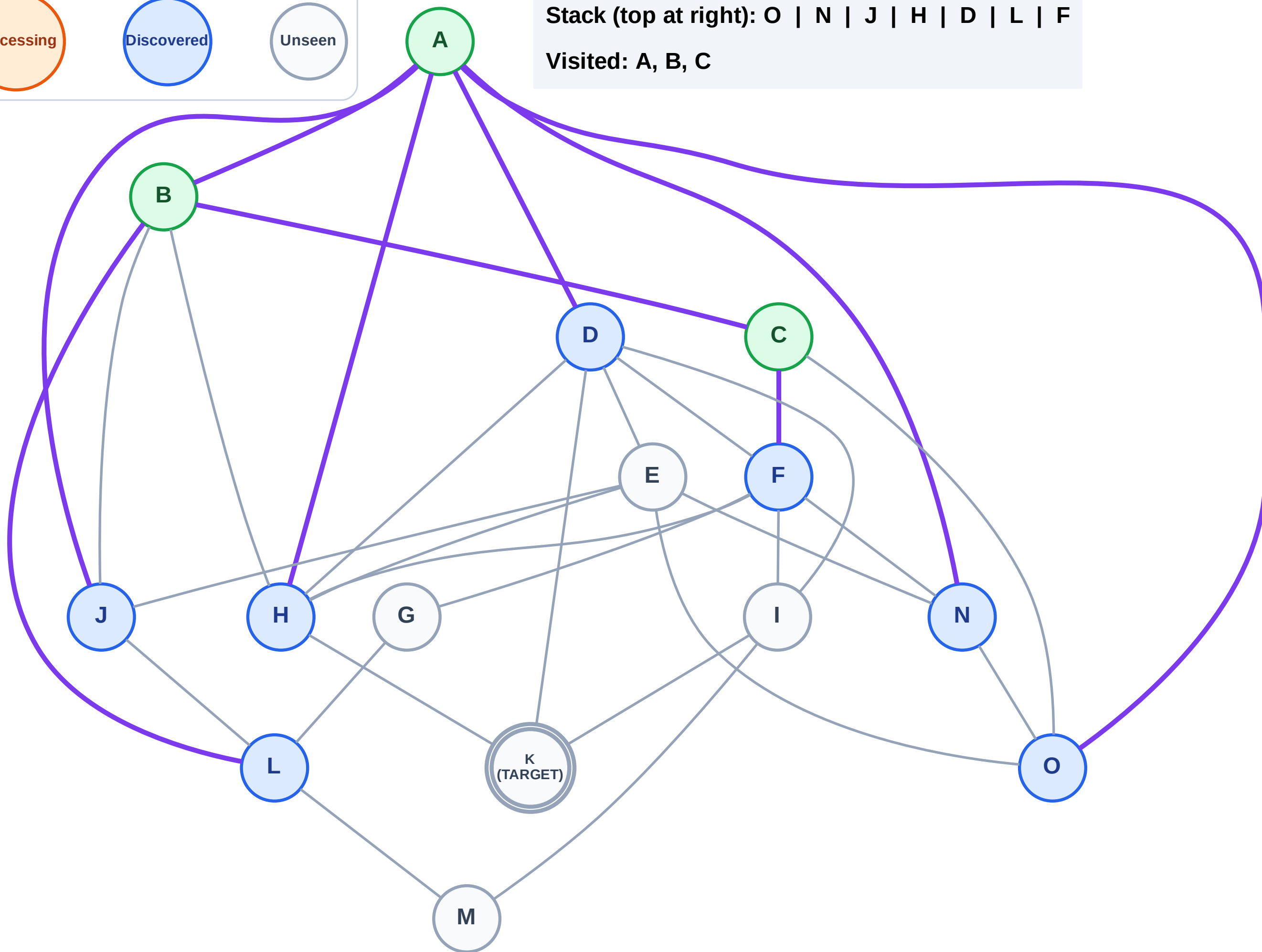
DFS Traversal

Step 7: Discover F from C



Stack (top at right): O | N | J | H | D | L | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

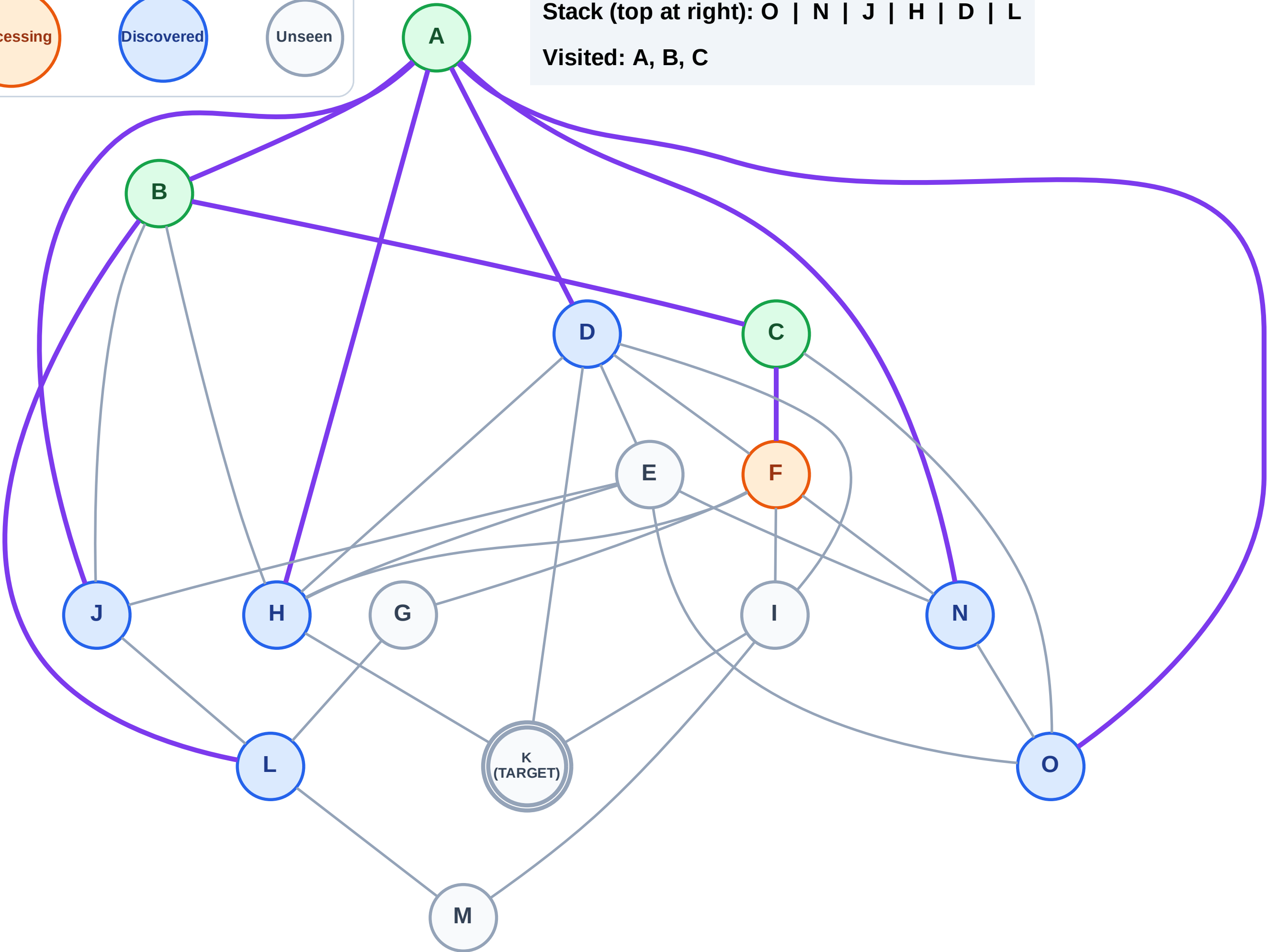
Processing

Discovered

Unseen

Stack (top at right): O | N | J | H | D | L

Visited: A, B, C



DFS Traversal

Step 9: Discover I, G from F

Legend

Complete

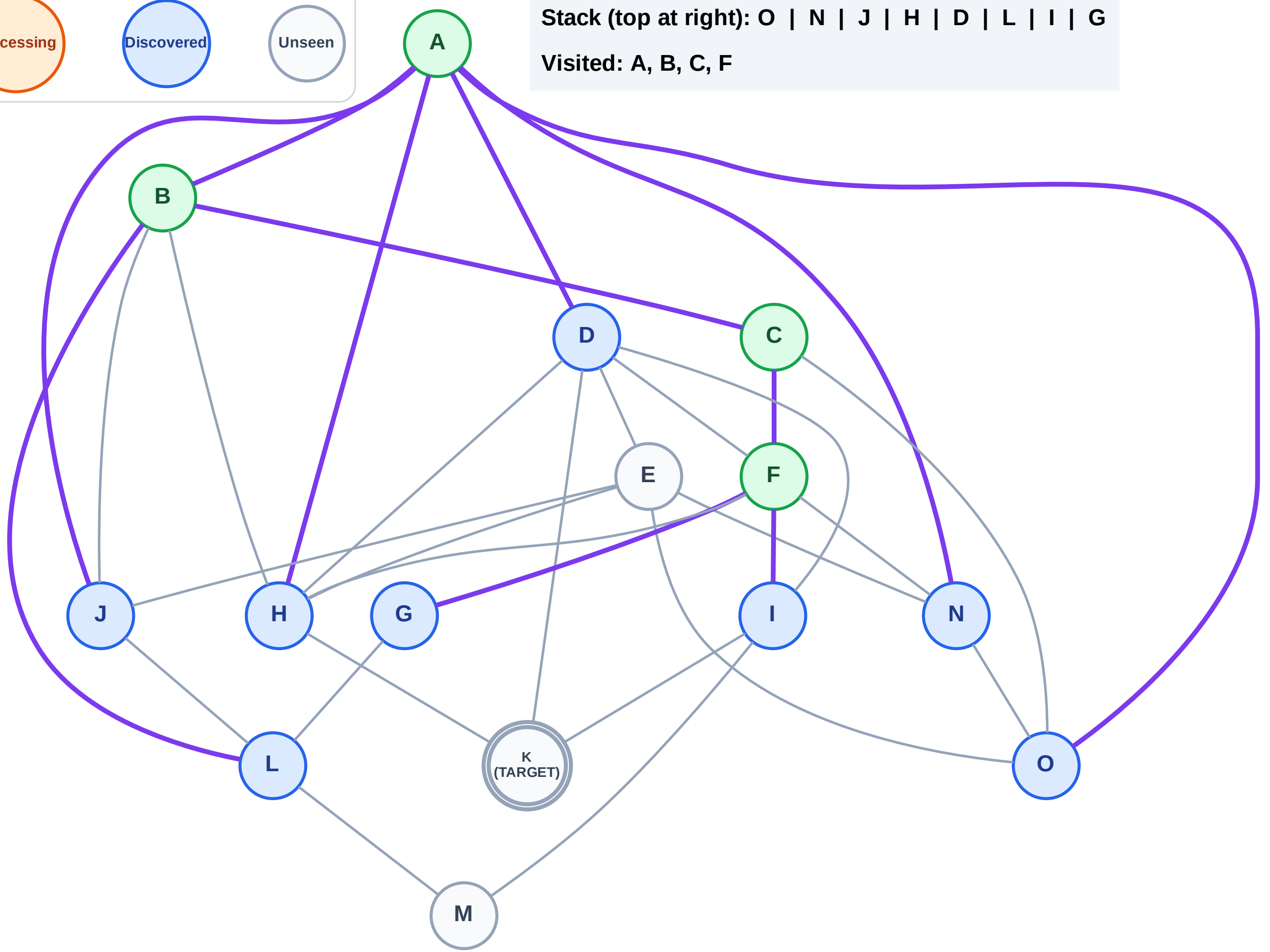
Processing

Discovered

Unseen

Stack (top at right): O | N | J | H | D | L | I | G

Visited: A, B, C, F



DFS Traversal

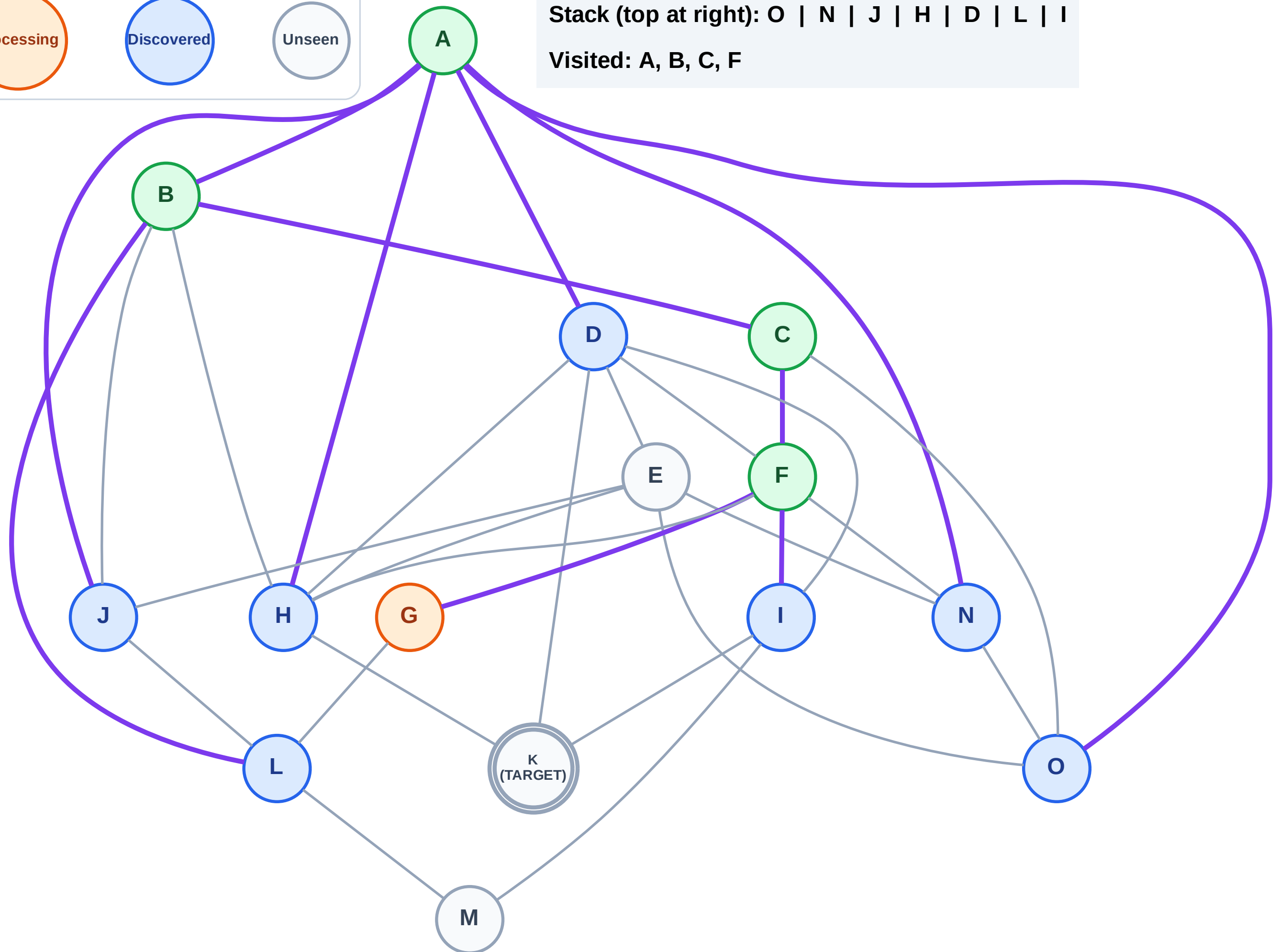
Step 10: Process node G

Legend



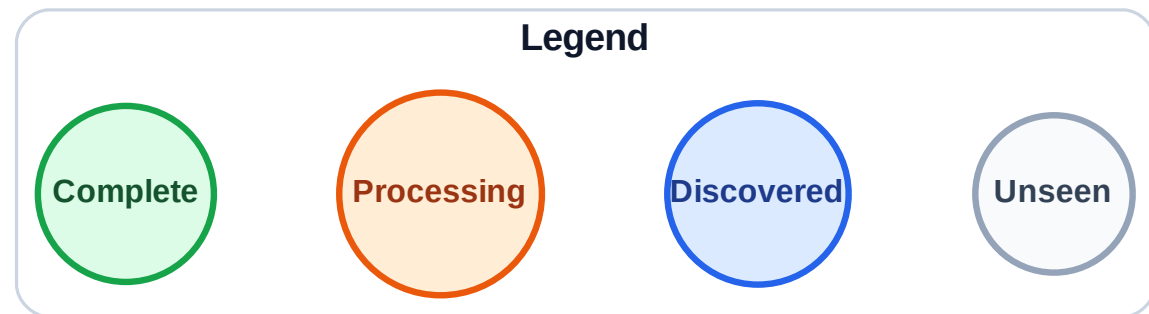
Stack (top at right): O | N | J | H | D | L | I

Visited: A, B, C, F



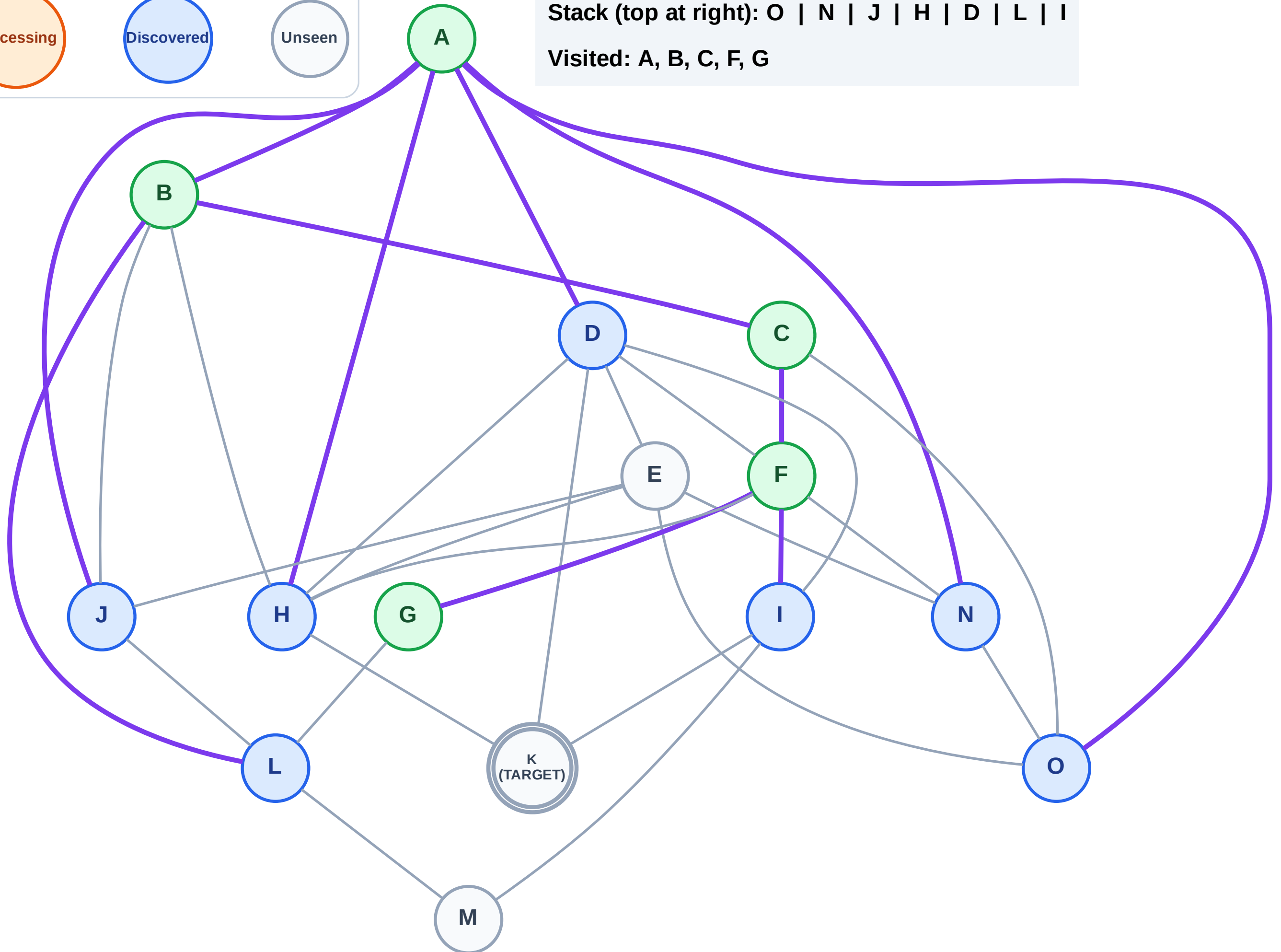
DFS Traversal

Step 11: No new neighbors from G



Stack (top at right): O | N | J | H | D | L | I

Visited: A, B, C, F, G



DFS Traversal

Step 12: Process node I

Legend

Complete

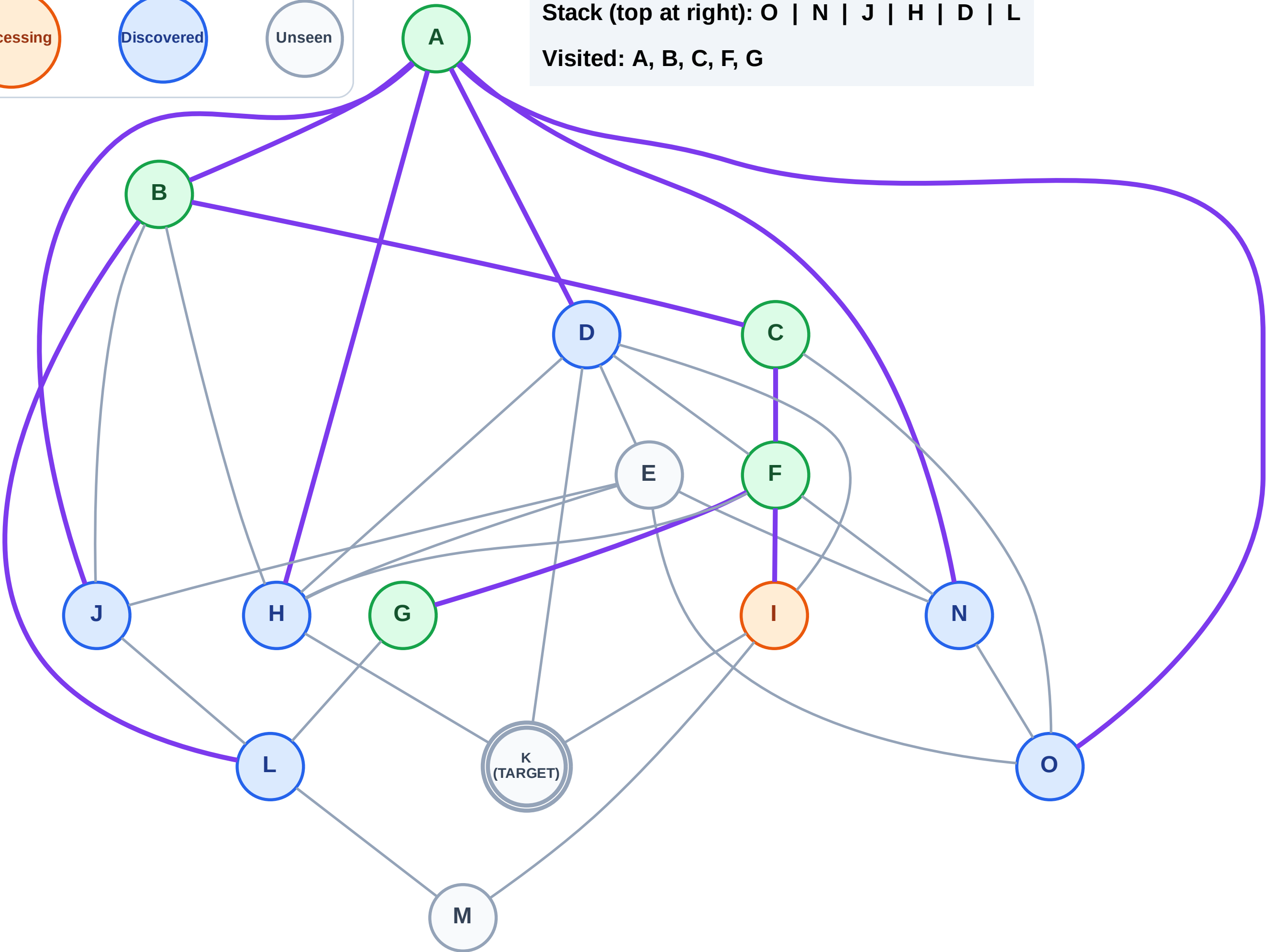
Processing

Discovered

Unseen

Stack (top at right): O | N | J | H | D | L

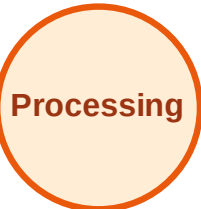
Visited: A, B, C, F, G



DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → C → F → I → K

Legend



Stack (top at right): O | N | J | H | D | L | M

Visited: A, B, C, F, G, I, K

